Social Research Methods · Group Project · Fudan University

The Psychology Behind Social Butterflies and Social Anxiety Among Fudan Undergraduates

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- Stratified sampling
- Scale development
- Factor analysis
- Nonresponse weighting
- Regression
- Mediation analysis

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01

Theoretical Framework

Background and research questions

Socializing is a big part of college life, and with the rise of social media, talk about social struggles has become common among students.

Social anxiety or social butterfly?

01 Do the two share similar psychological mechanisms?

02 Where do they overlap, and where do they differ?

03 How do these similarities and differences lead people to show different social patterns?

WHY IT MATTERS

Significance of the study

01

Understand

Get a picture of how college students socialize today.

02

Explain

Explore the psychological mechanisms behind students' social traits.

03

Guide

Socializing touches every part of student life and affects mental health. The findings can offer practical guidance for social and mental health education on campus.

LITERATURE

What existing research tells us

There is no empirical research yet on the “social butterfly” phenomenon, while social anxiety has been studied extensively. Over the past decade, many studies have looked at what drives social anxiety among college students:

Self-esteem

Liu et al., 2017; Zhang, Li & Yu, 2019; Han & Sun, 2015; Li & Wu, 2013

Parenting style

Li, Yang & Zhang, 2013

Fear of negative evaluation

Zhao & Dai, 2016; Peng et al., 2019

Self-focused attention

Li, Lu & Li, 2018

Personality traits

Wang, Huang & Wu, 2014; Shi et al., 2017

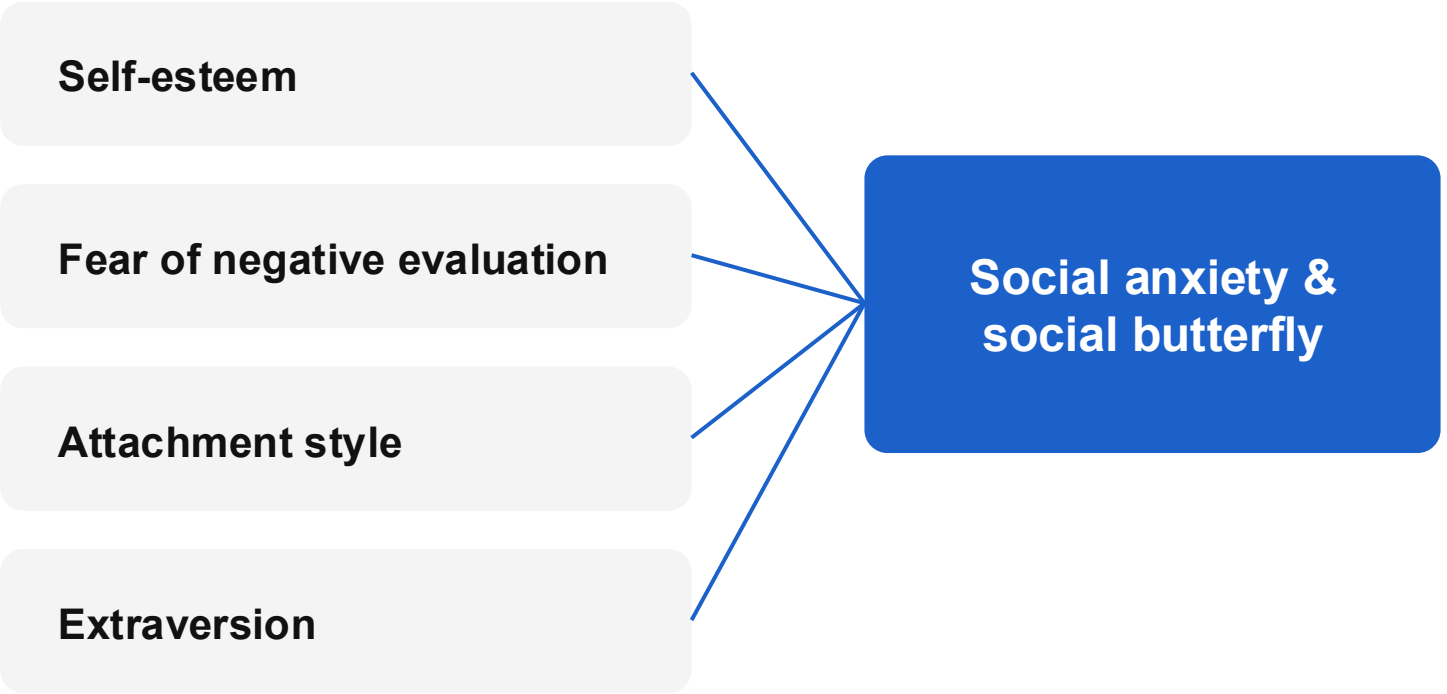
Attachment style

Wei & Huo, 2009

HYPOTHESES

Research questions and expected findings

Based on prior research, we chose four variables and examined how each relates to social anxiety and to being a social butterfly.



EXPECTED FINDINGS

- Self-esteem is negatively correlated with both social anxiety and being a social butterfly.
- Fear of negative evaluation mediates these links.
- Attachment style and extraversion moderate them.

02

Method

Sampling plan

120

Target sample size

6 variables × 20 = 120 completed surveys

400

Invitations sent

Sent to students' university email to allow for nonresponse

STRATIFIED SAMPLING: YEAR × DISCIPLINE (HUMANITIES & SOCIAL SCIENCES, STEM, MEDICINE)

1. Send 400 surveys to account for the expected response rate.
2. Allocate the 400 slots to each major in proportion to its enrollment, using official enrollment figures from the Office of Academic Affairs.
3. Assuming equal cohort sizes, split each major's slots evenly across years. Coverage error: if a cohort is overestimated, bounced emails reveal it; if underestimated, students with higher ID numbers may be left out of the frame.
4. Once each cell's size is fixed, draw students at random within the cell.

SAMPLING

Allocating the 400 slots across disciplines

Discipline	Majors in frame	Enrolled students	Population share	Slots allocated
Humanities & social sciences	10	5,641	37.5%	**150**
STEM	8	5,771	38.4%	**154**
Medicine	5	3,632	24.1%	**96**
Total	**23**	**15,044**	**100%**	**400**

Source: Fudan University Office of Academic Affairs enrollment by major. Recreated in English from the team's allocation spreadsheet.

HOW SLOTS WERE SPLIT

- Each major's slots = discipline slots × the major's share of enrollment, then split evenly across years and rounded.
- Two experimental classes in science and engineering overlap with other majors but admit similar numbers each year, so they were allocated 1:1.
- Medicine runs longer programs, so some cohorts (2014–2017) were sampled from combined pools.

How the survey was run

DISTRIBUTION

Surveys were emailed to students' university accounts, which made the invitation more credible.

HANDLING NONRESPONSE

1. Estimated the nonresponse rate and sent more invitations than the target sample.
2. Offered incentives: a personalized results report plus a lucky-draw cash prize. Because the report depends on their answers, respondents tend to answer more carefully.

INCENTIVE IN THE EMAIL

“After completing the survey, you will receive:

1. A personalized report, from a psychology perspective, on your social traits, how you compare with others, and the possible mechanisms behind them. We will email it in January.
2. A chance to win a ¥66 cash prize. Three people will be drawn at random.”

Limitation Although we used stratified sampling, offering a personalized report meant students more interested in psychology were more likely to respond, which biases the sample.

03

Results

SAMPLE

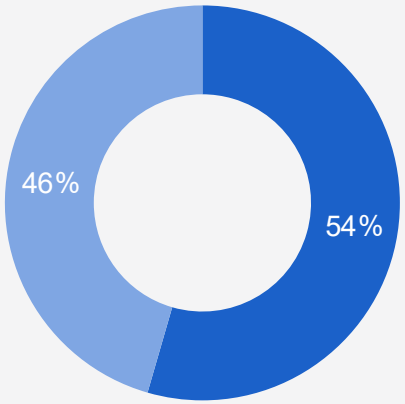
Who responded

35%

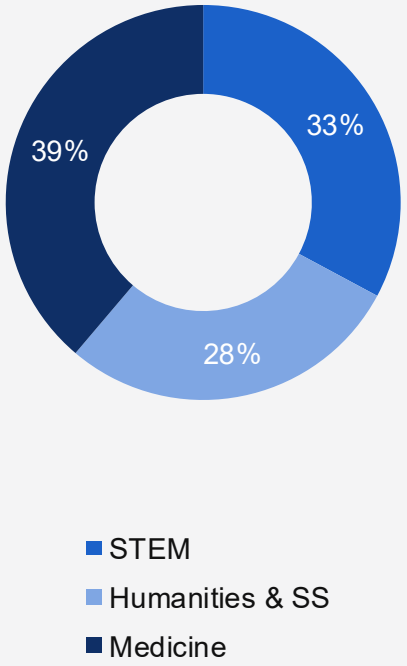
Response rate

134 valid responses

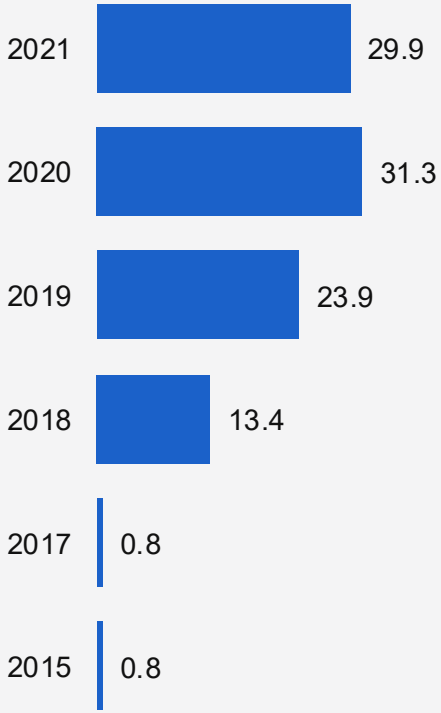
Gender



Discipline



Entry year (%)



2017 and 2015 entrants are medical students in longer programs.

Factor loadings for the new Social Butterfly scale

#	Item	Std. estimate
1	I often address others in a warm, familiar way	0.8353
2	I call out people's names loudly in public	0.8293
3	I often join other people's chats or discussions	0.8168
4	I can ask others for help without much hesitation	0.8288
5	I often liven up the mood in different settings	0.8152
6	I often invite friends out for meals	0.8350
7	I can express my thoughts without much hesitation	0.8355
8	I can start chatting with strangers I meet at school or work	0.8352
9	I don't feel awkward celebrating my birthday in public (e.g., at Haidilao)	0.8359
10	I speak up boldly when no one answers the teacher's question	0.8334
	Overall scale	**0.8445**

OTHER SCALES (OVERALL)

Fear of negative evaluation	**0.9415**
Self-esteem	**0.9124**
Social anxiety	**0.8654**
Attachment style	**0.7972**
Extraversion	**0.7234**

Weighting for nonresponse

Nonresponse made the sample's mix differ from the student population, so we weighted responses by discipline.

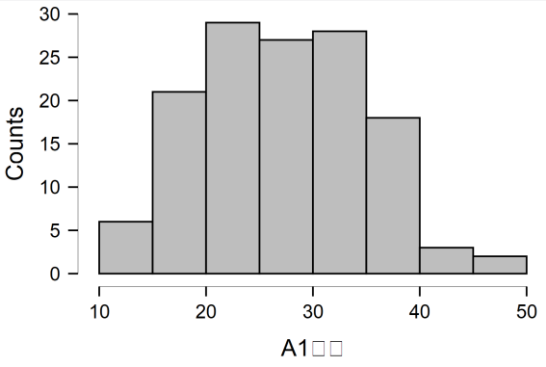
Discipline	Population share	Response rate	Nonresponse weight
Humanities & social sciences	0.375	0.253	**3.95**
STEM	0.385	0.338	**2.96**
Medicine	0.240	0.458	**2.18**

WHY NOT WEIGHT BY YEAR

1. After splitting by discipline, each year had only a few responses, so year weights would be unreliable.
2. Senior medical cohorts had no responses at all.

Score distributions

SOCIAL BUTTERFLY SCALE (MAX 50)

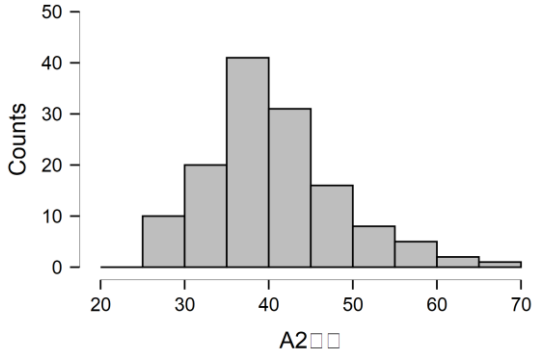


27.63

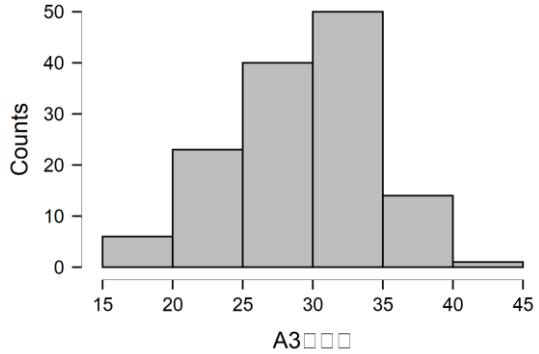
Sample mean

27.81 after weighting

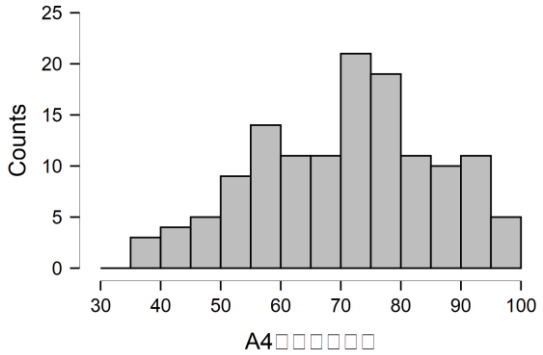
	STEM	Humanities & SS	Medicine
Sample size	52	38	44
Mean	25.788	**29.895**	27.864
SD	7.155	7.017	8.374
Min	10	14	12
Max	39	46	50



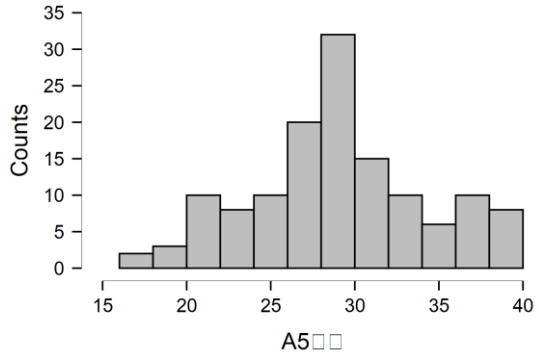
Social anxiety (A2)



Extraversion (A3)



Fear of negative evaluation (A4)



Self-esteem (A5)

Histograms exported from JASP; axis titles did not render.

Attachment styles

THREE DIMENSIONS OF THE ATTACHMENT SCALE

Closeness — Willingness to be physically and emotionally close

Dependence — Turning to attachment figures for support under stress or threat

Anxiety — Feeling anxious during long separations

2 (CLOSENESS-DEPENDENCE) × 2 (ANXIETY)

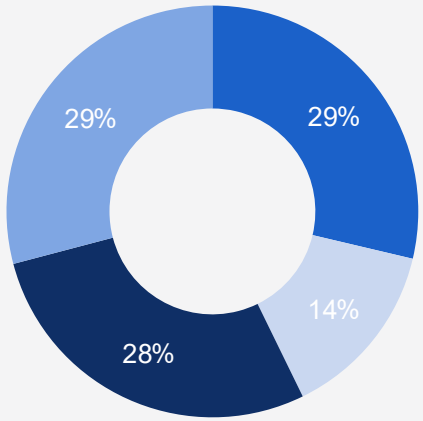
Secure
High closeness-dependence, low anxiety

Preoccupied
High closeness-dependence, high anxiety

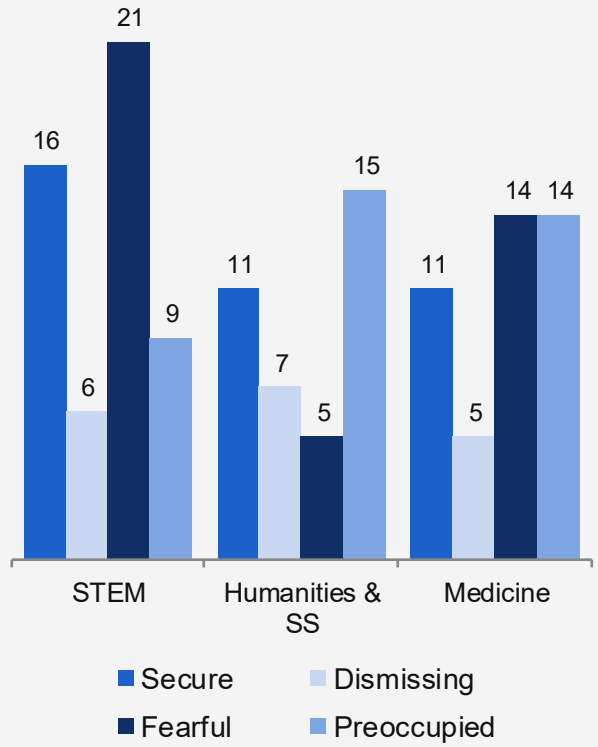
Dismissing
Low closeness-dependence, low anxiety

Fearful
Low closeness-dependence, high anxiety

Whole university (weighted)



Count by discipline



Correlation analysis

Pearson's Correlations

Variable		A1社牛	A2社恐
1. A1社牛	Pearson's r	—	
	p-value	—	
2. A2社恐	Pearson's r	0.385***	—
	p-value	< .001	—
3. A3外向性	Pearson's r	0.680***	0.456***
	p-value	< .001	< .001
4. A4惧怕负面评价	Pearson's r	−0.272**	−0.160
	p-value	0.001	0.065
5. A5自尊	Pearson's r	0.159	0.146
	p-value	0.067	0.093
6. A6依恋类型-亲近	Pearson's r	0.495***	0.258**
	p-value	< .001	0.003
7. A6依恋类型-依赖	Pearson's r	0.373***	0.251**
	p-value	< .001	0.003
8. A6依恋类型-焦虑	Pearson's r	−0.081	0.024
	p-value	0.349	0.782

* p < .05, ** p < .01, *** p < .001

KEY FINDINGS

Social anxiety and social butterfly are positively correlated.

Extraversion is significantly correlated with both social butterfly and social anxiety, p < .001.

Fear of negative evaluation is significantly correlated with social butterfly, p = .001.

Closeness and dependence are positively correlated with both social anxiety and social butterfly.

VARIABLE KEY

A1 Social butterfly · A2 Social anxiety · A3 Extraversion · A4 Fear of negative evaluation · A5 Self-esteem · A6 Attachment: closeness, dependence, anxiety

Social anxiety and social butterfly are the hypothesized outcome variables.

Multiple linear regression

Model Summary – A1社牛 ▼

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	7.668
H_1	0.708	0.501	0.493	5.458

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	3918.137	2	1959.068	65.755	< .001
	Residual	3902.946	131	29.793		
	Total	7821.082	133			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	27.634	0.662		41.715	< .001
H_1	(Intercept)	7.038	3.696		1.904	0.059
	A3外向性	0.938	0.089	0.658	10.588	< .001
	A4惧怕负面评价	−0.101	0.032	−0.198	−3.189	0.002

Model Summary – A2社恐

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	7.980
H_1	0.456	0.208	0.202	7.127

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	1763.346	1	1763.346	34.712	< .001
	Residual	6705.579	132	50.800		
	Total	8468.925	133			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	41.090	0.689		59.607	< .001
H_1	(Intercept)	20.979	3.468		6.049	< .001
	A3外向性	0.677	0.115	0.456	5.892	< .001

Social butterfly

Extraversion and fear of negative evaluation together significantly predict social butterfly, **$p < .001$, $R^2 = .501$**. Each predictor is also significant on its own.

Social anxiety

Extraversion significantly predicts social anxiety, **$p < .001$, $R^2 = .208$**.

Mediation model 1

Closeness and dependence mediate the link between fear of negative evaluation and social butterfly.

FEAR OF NEGATIVE EVALUATION → CLOSENESS → SOCIAL BUTTERFLY

Results

Mediation Analysis

Parameter estimates

Direct effects

			Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	Upper
惧怕负面评价	→	社牛	−0.086	0.039	−2.215	0.027	−0.162	−0.010

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Indirect effects

				Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	Upper
惧怕负面评价	→	依恋类型-亲近	→ 社牛	−0.053	0.022	−2.475	0.013	−0.096	−0.011

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Total effects

			Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	Upper
惧怕负面评价	→	社牛	−0.139	0.043	−3.271	0.001	−0.223	−0.056

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

FEAR OF NEGATIVE EVALUATION → DEPENDENCE → SOCIAL BUTTERFLY

Results

Mediation Analysis

Parameter estimates

Direct effects

			Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	Upper
惧怕负面评价	→	社牛	−0.104	0.041	−2.524	0.012	−0.184	−0.023

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Indirect effects

				Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	Upper
惧怕负面评价	→	依恋类型-依赖	→ 社牛	−0.036	0.017	−2.138	0.033	−0.068	−0.003

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Total effects

			Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	Upper
惧怕负面评价	→	社牛	−0.139	0.043	−3.271	0.001	−0.223	−0.056

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Mediation model 2

Dependence mediates the link between extraversion and social butterfly.

Extraversion → Dependence → Social butterfly

Results

Mediation Analysis

Parameter estimates

Direct effects

						95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
外向性	→	社牛	0.894	0.095	9.406	< .001	0.708	1.080

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Indirect effects

								95% Confidence Interval	
					Estimate	Std. Error	z-value	p	Lower	Upper
外向性	→	依恋类型-依赖	→	社牛	0.076	0.038	1.987	0.047	0.001	0.150

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Total effects

						95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
外向性	→	社牛	0.970	0.090	10.732	< .001	0.793	1.147

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

04

Conclusions & Discussion

Main conclusions

01

Social anxiety and social butterfly are correlated.

02

Extraversion (+) and fear of negative evaluation (−) predict social butterfly; extraversion (+) predicts social anxiety.

03

Fear of negative evaluation (−) affects social butterfly directly, and indirectly through dependence (+) and closeness (+).

04

Extraversion affects social butterfly directly (+), and indirectly through dependence (+).

Discussion and next steps

FUTURE RESEARCH

Why are more extraverted people more socially anxious?

Extraversion shapes social butterfly through attachment, but how it drives social anxiety is still unclear.

Why are social anxiety and social butterfly correlated?

They may share similar mechanisms, but we need to explore more variables.

Are they two ends of one spectrum, or can they coexist?

WHY THE HYPOTHESES WEREN'T SUPPORTED

- Little prior research to build on.
- Social anxiety and social butterfly are correlated, but social butterflies are not afraid of negative evaluation, which overturned our hypothesis.

CONTRIBUTION

We built our own Social Butterfly scale and showed it is highly consistent. Turning a cultural concept into a measurable variable is new ground.

05

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